Adaptive Agent Modes v2 - Candidate Report

Status: the six-call Candidate screen completed with execution integrity but is comparatively invalid. Two harness confounders were repaired with zero-provider tests. No replacement calls, Stable change or promotion are authorized; this is not Stable guidance.

Executive result

The previous Vanilla-versus-Full sentinel did not show that agentic guidance was useless. Two shared failures came from invalid evaluation contracts: whitespace-only input was hidden behind the phrase "non-empty", and semantically correct documentation was rejected by a literal substring. The only unaffected pair scored 100/100 and was a ceiling. The correct response was to freeze the conclusion, repair evaluation validity and make components independently measurable.

Candidate v2 implements five runtime modes. Runtime Vanilla is advisory/read-only; coding benchmarks instead use an unscaffolded task-only `vanilla-control`. Selection uses three dimensions: consequence is the only path to Full, while breadth and continuity cap at Mode 3. Planning a critical action is distinct from executing it. The policy chooses the lowest mode covering the evidenced dimensions and escalates on scope growth, ambiguity, repeated verification failure, critical action or session boundary. Numeric ceilings remain hypotheses pending benchmark calibration.

What changed

Router precision

The old router selected research rules because an API response contained `source`, security because a schema contained `email`, and migrations because a parser returned an `index`. V2 requires strong phrases or contextual co-occurrence and supports negative matches. All 16 mode cases pass, including the known false-positive and action-intent traps.

Objective integrity and enforcement

The objective compiler covers explicit list items, the task objective and normative prose outside code fences. Every material requirement gets a stable ID and evidence kinds. Ambiguous wording such as non-empty versus non-blank, or semantic versus verbatim documentation, is recorded before implementation.

Completion is fail-closed. Mode 2 and above require every applicable ledger row to have reproducible evidence, material ambiguities to be authoritatively resolved, verification commands to pass and protected inputs to retain their hashes. Full also needs human scope approval and an independent verifier PASS. Hidden graders remain isolated, host-only evaluation evidence and cannot be used as an agent's completion gate.

Adaptive cost and model policy

Modes select durable capability profiles, not product names. The current weekly provider catalog resolves the profile and records the actual model after execution. Expiration blocks automatic selection and creates a model-refresh Candidate; it does not invent an ID or silently alter the user's default.

The first scaffolding screen uses one validated open-world parser fixture and the same resolved model/effort for every arm. It costs six provider calls, not five: five solutions plus the independent Full verifier. There are no replacements, automatic follow-ups or paid calls without a separate approval.

Evidence

- Candidate unit tests: 12/12.
- Candidate readiness gates: 8/9 PASS; capability activation deliberately FAILS closed.
- Router mode cases: 16/16 (including breadth-cap, consequence-beats-breadth and action-intent regressions: wide mechanical work stays at Mode 3, an executable one-line credential rotation selects Full, while explanation and runbook-only requests do not).

- Real-world fixture validity: 6/6; every reference scores 100.
- Negative controls: 12/12 pass public tests and are rejected by hidden graders.
- Semantic documentation variants: both formerly rejected agent solutions now score 100 under the versioned semantic oracle.
- Adaptive workspace preflight: 5/5 arms prepared, zero hidden paths, zero provider calls.
- Historical classification: execution integrity PASS; affected quality checks UNINTERPRETABLE; original records unchanged.

First outcome screen

Campaign `20260714-111231-adaptive-mode-ablation` executed exactly six sequential Codex calls on `gpt-5.6-sol` at medium effort: five solution arms and one independent Full verifier. There were no replacements or automatic follow-ups. Subscription CLI monetary cost was not reported.

Arm	Product score	Wall seconds	Observed tokens	Validity
Vanilla control	25	26.6	38,259	Invalid control prompt
Mode 1 Lean	100	187.7	388,416	Valid but ceiling
Mode 2 Routed	100	225.8	419,491	Valid but ceiling
Mode 3 Assured	100	217.9	421,147	Valid but ceiling
Full plus verifier	100	326.1	576,757	Product and verifier PASS; host gate invalid

The screen cannot support a mode recommendation. Runtime Vanilla had recently become advisory/read-only, but the benchmark mistakenly reused that preset and explicitly prohibited the supposed coding control from editing; it changed no files. Full produced a 100-point solution and its independent verifier passed, but the Windows host finalizer treated the verifier's Linux `python3` launcher as a product failure (exit 9009). Mode 1 through Mode 3 all reached a 100-point ceiling, so they do not discriminate adjacent boundaries.

The historical records remain unchanged. The derivative summary is `candidate-screen-invalid`, its raw Mode 1 recommendation is suppressed, and Stable remains untouched. Candidate now prepares Vanilla as a distinct unscaffolded task-only control and resolves a leading `python3` command to the pinned host interpreter when necessary. Both repairs pass zero-provider regression tests.

Capability activation audit

A second fail-closed audit found that several declared mode differences are not yet executable. Mode 3 names impact mapping, adversarial verification, context telemetry, durable checkpoints and session handoff, but the current compiler does not create or gate most of those artifacts. Mode 1 names a compact requirement ledger but currently receives the same full ledger representation as higher modes. Seven of fourteen declared capabilities are active; the remainder are missing or partial. Candidate readiness is therefore FAIL even after repairing the two campaign confounders. No replacement benchmark should run until this gate passes.

Expected impact

Mode 1 avoids heavyweight scaffolding for small reversible work. Mode 2 makes ordinary implementation accountable to the complete objective. Mode 3 adds impact and adversarial controls where hardcoding, stale consumers, misleading green tests and failure boundaries are likely; it is also the ceiling for breadth-only and continuity-only escalation, so wide mechanical changes and multi-session implementations land here with durable checkpoints instead of paying for Full. Full reserves the human gate and independent verifier for critical consequence alone: outward-facing actions, credential rotation, coordinated production data rollouts and cross-system releases. Size never buys the human gate; blast radius does.

This is a structural and validity improvement, not yet proof of production quality lift. First implement and verify every declared capability. Only then design a replacement fixture that separates Mode 1, Mode 2 and Mode 3; any provider screen requires new explicit approval. Promotion remains blocked until a valid replicated outcome exists.

Sources used

Primary and empirical	Platform and safety
Evaluating AGENTS.md	Anthropic: context engineering
Probe-and-Refine guidance	Anthropic: long-running harnesses
ContextBench	OpenAI Codex documentation
METR productivity study	Claude Code documentation
SWE-bench	NIST SP 800-218A
OpenHands Index	OWASP MCP Security

YouTube, podcasts and social/engineering feeds remain research radar inputs in the persistent source registry; they do not override primary evidence or local evals.